MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6533WA

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Triode, subminiature, high Mu

Outline	---	3-1 (EIA)
Base	---	E8-10
Envelope	---	T3
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	a	g	g	a	k	h	h	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rk	Rg	Ib	Pp	TE	Alt
Unit:	V	V dc	V dc	v	Ohms	Meg	mA dc	W	°C	ft
Maximum:	6.9	150	0, -55	200	---	1.2	2.5	0.35	220	Note 1
Minimum:	5.7	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	120	0	0	1,500	---	---	---	---	---

GENERAL:

Qualification - Required

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1								
1301	Heater current			0.4	II	If	190	210	mA dc
1336	Heater-cathode leakage			0.4	II	Ihk	---	5	µA dc
1266	Total grid current	2		0.4	II	Ic	0	-0.1	µA dc
1256	Electrode current (1) (anode)	2		0.4	II	Ib	0.6	1.25	mA dc
1256	Electrode current (2) (anode)		Ec1 = -3.5 V dc	0.4	II	Ib	---	50	µA dc
1306	Transconductance (1)	2		0.4	II	Sm	1,400	2,100	µmhos
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes			2.5	S3	---	---	---	---
1256	Electrode current (3) (anode)		Ec1 = -2.0 V dc	2.5	I	Ib	5	---	µA dc
1306	Transconductance (2)		Ef = 5.7 V	2.5	I	ΔSm Ef	---	7.5	%
1266	Grid emission	3	Ec1 = -3.5 V dc	---	---	Ic	0	-0.3	µA dc
1246	Audio frequency noise	4	Esig = 35 mV ac; Rg = 1.0 Meg; Rk = 1,500 ohms; Rp = 0.2 Meg	2.5	I	---	---	---	---
1316	Amplification factor			---	---	Mu	48	60	---
1331	Direct-interelectrode capacitance		No shield	6.5	Code E	Cgp	1.2	2.0	pF
						Cin	1.3	2.2	pF
						Cout	0.4	0.8	pF
1256	Electrode current (1) (anode) (tube operation time)	5	t = 20	4.0	S3	Ib(1)	90	110	%
1031	High-frequency vibration		Rp = 10,000 ohms	2.5	I	Ep	---	1.0	mV ac
1031	Sweep-frequency vibration		Rp = 10,000 ohms; F = 30 to 1,000 Hz; 15 G; t = 3 min; positions X and Y only	4.0	S3	ep	---	15	mv (peak to peak)

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 2 -Continued								
1116	Lead fatigue			2.5	Code G	---	---	---	---
1041	Shock	6	450 G; Ehk = 100 V dc	6.5	See note 7	---	---	---	---
1031	Vibration-fatigue (1)			6.5	See note 7	---	---	---	---
1031	Vibration-fatigue (2)		10 G; t = 6 hours	6.5	See note 7	---	---	---	---
---	Shock and vibration-fatigue (1) and (2)-test end points								
1031	High-frequency vibration		Rp = 10,000 ohms	---	---	Ep	---	3.0	mV ac
1336	Heater-cathode leakage			---	---	Ihk	---	10	µA dc
1306	Change in transconductance (1) of individual tubes			---	---	ΔSm_t	---	10	%
1266	Total grid current			---	---	Ic	0	-1.0	µA dc
2126	Envelope strain			6.5	I	---	---	---	---
1105	Permanence of marking			---	---	---	---	---	---

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Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.5 V; Ehk = 140 V ac; Ec1 = 0; Eb = 0	---	---	---	---
---	Heater-cycling life-test end point:						
1336	Heater-cathode leakage			Ihk	---	20	µA dc
1516	Stability life		Eb = 150 V dc; Rk = 680 ohms; Ehk = +200 V dc; Rg1 = 1.0 Meg; TA = room	---	---	---	---
---	Stability life-test end point: (1 hour)						
1306	Change in transconductance (1) of individual tubes			ΔSm t	---	10	%
1501	Intermittent life	8,9	Eb = 150 V dc; Rk = 680 ohms; Ehk = +200 V dc; Rg1 = 1.0 Meg; TE = 220°C (min)	---	---	---	---
---	Intermittent life-test end points (500 hours):	10					
1266	Total grid current			Ic	0	-0.5	µA dc
1301	Heater current			If	185	215	mA
1306	Change in transconductance (1) of individual tubes			ΔSm t	---	20	%
1306	Transconductance (2)		Ef = 5.7 V	ΔSm Ef	---	15	%
1336	Heater-cathode leakage			Ihk	---	10	µA dc
1211	Insulation of electrodes			---	---	---	---
1306	Transconductance (1), average change			Avg ΔSm t	---	15	%

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Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 3 -Continued						
---	Intermittent life-test end points (1,000 hours)	10					
1266	Total grid current			Ic	0	=1.0	µA dc
1301	Heater current			If	177	223	mA
1306	Change in transconductance (1) of individual tubes			ΔSm t	---	30	%
1336	Heater-cathode leakage			Ihk	---	15	µA dc

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. This test shall be performed at the conclusion of the holding period.
3. Prior to this test, tubes shall be preheated a minimum of 5 minutes operating at the conditions indicated below. The 3-minute test shall not be permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid emission test.

Ef V	Ec1 V dc	Eb V dc	Rk Ohms	Rg Meg
7.5	0	120	1,500	1.0

4. The rejection level shall be set at the VU meter reading obtained during calibration.
5. Tube operation time is the time in seconds required for the anode current to attain a value within ±10 percent of the 3-minute anode current value when measured at the test conditions specified for electrode current (1) (anode).
6. A grid resistor of 0.1 megohm shall be added, except when a thyratron-type short indicator is used.
7. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code E, shall apply.
8. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.
9. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples and shall not exceed a total of three tube failures.
10. The allowable defectives per characteristic shall not be greater than 1 for the first sample and shall not be greater than 3 for combined samples.
11. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 17, 99
DLA - ES

User activities:
Army - AR
Navy - AS, CG, MC, SH, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5960-3299)

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